NORTHERN UNIVERSITY
OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Technology

Software requirements specification (SRS)

KrishiBondhu AI – An Intelligent and Personalized Agricultural Decision Support System

Course Code: CSE 4104
Course Title: Software Development III

Submitted To

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1. Introduction

1.1 Project Title

KrishiBondhu AI - Smart Farming Assistant: a Bangla-friendly, AI-powered agricultural assistance platform for Bangladesh.

1.2 Project Overview

KrishiBondhu AI ("Krishi Bondhu" means "Farmer's Friend") is a web application that gives farmers in Bangladesh easy access to reliable agricultural information and AI-assisted guidance in both Bangla and English. Users can ask farming questions to an AI assistant, upload crop images for a preliminary disease analysis, receive fertilizer and crop recommendations, check weather updates, learn about common pests, keep farm records, set farming reminders and save recommendations for later use. The AI features are powered by the Google Gemini API through a Node.js/Express backend, with a React frontend and a MongoDB database.

1.3 Problem Statement

Farmers in Bangladesh often find it difficult to access reliable agricultural information and to find appropriate solutions to farming problems such as crop diseases, pest attacks, fertilizer use and crop selection. Access to agricultural experts is limited, particularly in rural areas, and language barriers make many digital resources hard to use. As a result, farmers may rely on guesswork, delayed advice or non-specific information, which can reduce crop yield and increase costs. There is a need for an accessible, Bangla-friendly platform that provides timely, understandable and practical agricultural guidance.

1.4 Objectives

- Provide agricultural information and AI assistance in Bangla and English.
- Help farmers identify and understand common crop-related problems, including preliminary crop disease detection from images.
- Provide AI-assisted farming recommendations covering crop selection, fertilizer use and pest management.
- Improve access to agricultural knowledge for farmers, agricultural students, advisors and rural communities.
- Support better agricultural decision-making through weather updates, farm records, reminders and saved recommendations.

1.5 Scope of the Project

In scope

- User registration, secure login and profile management with Bangla/English language support.
- AI agricultural assistant (text chat) powered by the Google Gemini API.
- Crop disease detection (preliminary, image-based) and pest management information.
- Fertilizer guidance and crop recommendations based on soil, season and location.
- Weather information for the user's location.
- Farm records, farming reminders and saved recommendations.
- Advisor features (review flagged AI responses, maintain knowledge entries) and Admin features (user management, monitoring, reports).

- A responsive web application accessible from desktop and mobile browsers.

Out of scope (for the current release)

- Native mobile applications (iOS / Android).
- Online payments, e-commerce or marketplace features.
- Voice input and SMS-based notifications.
- Offline operation of AI features.
- Replacing professional agricultural experts: AI output is a preliminary, advisory aid only.

1.6 Target Users

- Farmers in Bangladesh
- Agricultural students
- Agricultural advisors
- Rural communities

1.7 Proposed Technology Stack

Layer	Technology
Frontend	React (single-page web application, responsive design)
Backend	Node.js with Express (RESTful API, JWT-based authentication)
Database	MongoDB (document database)
AI Integration	Google Gemini API (text chat and image analysis)
External Data	Third-party weather API (e.g. OpenWeatherMap)
Email	SMTP email service (e.g. Nodemailer) for password-reset emails
Version Control	Git and GitHub
Deployment	To be finalised in the development phase

1.8 Definitions, Acronyms and Abbreviations

Term	Meaning
SRS	Software Requirements Specification
FR / NFR	Functional Requirement / Non-Functional Requirement
AI	Artificial Intelligence
Gemini API	Google's generative AI service used for text and image understanding
API / REST	Application Programming Interface / Representational State Transfer
JWT	JSON Web Token, used for stateless authentication
RBAC	Role-Based Access Control
SPA	Single-Page Application
UC	Use Case
ER	Entity Relationship

1.9 Assumptions and Constraints

- Users have a device with a web browser and an internet connection (mobile data is sufficient).
- AI and weather features depend on the availability and usage limits of the Google Gemini API and the weather API; password reset depends on an email service.
- AI-generated results (including disease detection and fertilizer advice) are preliminary and advisory; accuracy cannot be guaranteed.
- The system uses only the data entered by the user and the data provided by external services; it does not read soil or field sensors.

- The project is developed within the CSE 4104 course timeline by a three-member team using free or student-tier services where possible.

2. Functional Requirements

Functional requirements describe what the system will do. Priority: **H = High, M = Medium, L = Low.**

2.1 Authentication and Account Management

ID	Requirement	Priority
FR-01	Users can register an account using their name, email address and password (a mobile number can be added optionally).	H
FR-02	The system validates registration data (unique email, valid format, password strength) and shows clear error messages in the user's chosen language.	H
FR-03	Users can log in securely using their registered credentials.	H
FR-04	The system maintains authenticated sessions that expire after a period of time, and users can log out.	H
FR-05	Users can reset a forgotten password through an email-based reset link.	M

2.2 Profile and Language

ID	Requirement	Priority
FR-06	Users can view and update their profile (name, contact details, district, upazila, main crops and farm size).	H
FR-07	The system provides the interface in Bangla and English, and users can switch language at any time.	H
FR-08	The system remembers each user's preferred language.	M

2.3 AI Agricultural Assistant

ID	Requirement	Priority
FR-09	Users can ask farming-related questions to the AI assistant in Bangla or English text.	H
FR-10	The system sends the question, together with relevant profile context (location, crops), to the Gemini API and displays the generated answer in the language of the question.	H
FR-11	The system stores chat sessions and messages for each user.	H
FR-12	Users can view their previous chat sessions and delete any session.	M
FR-13	Users can rate an AI response as helpful or not helpful and flag a response as incorrect or unsafe.	M
FR-14	The system displays a notice that AI responses are advisory and not a substitute for expert consultation.	H

2.4 Crop Disease Detection

ID	Requirement	Priority
FR-15	Users can upload a crop image (JPG or PNG) from their device or camera.	H
FR-16	The system validates the image type and size before accepting the upload.	H

FR-17	The system analyses the image using the Gemini API and returns a preliminary result: probable disease or problem, confidence level, symptom explanation and suggested management steps.	H
FR-18	If the image is unclear or does not show a crop, the system informs the user and asks for a better image.	M
FR-19	The system saves each scan result (image, result, date) in the user's history and optionally links it to a farm record.	M

2.5 Fertilizer Guidance

ID	Requirement	Priority
FR-20	Users can enter crop, soil type, growth stage and field area to request fertilizer guidance.	H
FR-21	The system generates general fertilizer recommendations (type, approximate quantity guidance and timing) with safety notes.	H

2.6 Crop Recommendation

ID	Requirement	Priority
FR-22	Users can provide soil type, season and location (or use their profile location) to request crop suggestions.	H
FR-23	The system returns suggested crops with short reasons (suitability, water requirement, growth duration).	H

2.7 Weather Updates

ID	Requirement	Priority
FR-24	The system shows current weather and a short-term forecast for the user's location.	H
FR-25	The system displays farming-related weather advisories (for example, a heavy-rain warning that may affect irrigation or spraying decisions).	M

2.8 Pest Management

ID	Requirement	Priority
FR-26	Users can browse and search a pest library by name (Bangla or English) or by affected crop.	H
FR-27	The system shows pest details: description, symptoms, affected crops and management methods.	H

2.9 Farm Records

ID	Requirement	Priority
FR-28	Users can create farm/crop records (crop, field name, area, soil type, sowing date and expected harvest date).	H
FR-29	Users can view, search/filter, edit and delete their farm records.	H
FR-30	Users can add dated notes to a farm record.	M

2.10 Reminders

ID	Requirement	Priority
FR-31	Users can create reminders for farming activities (irrigation, fertilizing, spraying, harvesting) with a date and time, optionally linked to a farm record.	H
FR-32	Users can edit, delete and mark reminders as completed.	M
FR-33	The system notifies users in the application when a reminder is due and shows upcoming and overdue reminders on the dashboard.	H

2.11 Saved Recommendations

ID	Requirement	Priority
FR-34	Users can save any AI recommendation (chat answer, disease result, fertilizer or crop recommendation).	M
FR-35	Users can view, search and delete their saved recommendations.	M

2.12 Advisor Functions

ID	Requirement	Priority
FR-36	Advisors can view AI responses that users have flagged.	M
FR-37	Advisors can submit corrections or verified guidance and mark a flagged response as reviewed.	M
FR-38	Advisors can add and update pest and crop knowledge entries.	M

2.13 Admin Functions

ID	Requirement	Priority
FR-39	Admins can view, search, suspend, reactivate and delete user accounts.	H
FR-40	Admins can assign roles (Farmer/User, Advisor, Admin) to accounts.	H
FR-41	Admins can manage the pest and crop knowledge base.	M
FR-42	Admins can monitor system activity through activity logs.	M
FR-43	Admins can generate usage reports (for example, users, AI queries and disease scans) filtered by date range.	M

2.14 General System Functions

ID	Requirement	Priority
FR-44	The system stores all user, farm, chat, scan, reminder and recommendation data in the database.	H
FR-45	If an external service (Gemini API or weather API) is unavailable, the system shows a clear message and lets the user retry later, while other features continue to work.	H
FR-46	The system records important actions (login, account deletion, role change, flagged-response review) in an activity log.	M

3. Non-Functional Requirements

Non-functional requirements describe the quality, performance and constraint expectations of the system.

3.1 Performance

ID	Requirement
NFR-01	Pages shall load within 3 seconds on a standard broadband connection and within 5 seconds on a typical 4G mobile connection.
NFR-02	Non-AI API requests (login, records, reminders, pest search) shall respond within 2 seconds for at least 95% of requests under normal load.
NFR-03	AI chat answers shall be returned within 10 seconds and crop image analysis within 15 seconds under normal conditions, with a visible loading indicator.
NFR-04	Uploaded images shall be compressed or resized on the client before upload to reduce bandwidth use.

3.2 Usability and Localization

ID	Requirement
NFR-05	The system shall provide a responsive user interface that works on desktop, tablet and mobile devices.
NFR-06	The interface shall be simple, with icons, large touch targets and short plain-language text suitable for users with limited digital literacy.
NFR-07	Bangla text shall render correctly (Unicode with a Bengali-supporting font) throughout the application, including AI responses.
NFR-08	The main features (ask the AI assistant, scan a crop image) shall be reachable within two taps/clicks from the home screen.

3.3 Security

ID	Requirement
NFR-09	User passwords shall be securely hashed and salted (for example, bcrypt) and never stored or logged in plain text.
NFR-10	Authentication shall use signed tokens (JWT) with an expiry time.
NFR-11	All communication between client and server shall use HTTPS in the deployed system.
NFR-12	Role-based access control shall ensure that Farmer, Advisor and Admin users can access only the functions permitted to their role.
NFR-13	All user input shall be validated and sanitized on the server to prevent injection (including NoSQL injection) and cross-site scripting attacks.
NFR-14	API keys (for example, the Gemini API key) shall be stored in server-side environment variables and shall never be exposed to the client or committed to the GitHub repository.
NFR-15	The login and AI endpoints shall use rate limiting to reduce brute-force attempts and abuse.

3.4 Data Privacy and Protection

ID	Requirement
NFR-16	Personal data, farm records, chat history and images shall be accessible only to their owner and authorized staff, and only the minimum data needed (never passwords or contact details) shall be sent to external AI services.
NFR-17	Users shall be able to delete their account and associated data.
NFR-18	Uploaded crop images shall be stored with access restricted to the owning user.

3.5 Scalability

ID	Requirement
NFR-19	The system shall support at least 100 concurrent users in the initial deployment.
NFR-20	The backend shall be stateless and modular so that it can be scaled horizontally and new features can be added without major redesign.

3.6 Availability and Reliability

ID	Requirement
NFR-21	The system shall target high availability (at least 99% during the demonstration period), subject to the limits of the chosen hosting platform.
NFR-22	The system shall degrade gracefully when an external service is unavailable: only the affected feature is disabled and a clear message is displayed.
NFR-23	The database shall be backed up regularly (at least daily) so that data can be restored after a failure.

3.7 AI Quality and Safety

ID	Requirement
NFR-24	Every AI-generated recommendation shall be accompanied by a notice that it is advisory and should be confirmed with a local agricultural expert.
NFR-25	AI prompts shall be designed to keep answers within the agricultural domain, to respond in the user's language and to avoid unsafe guidance.
NFR-26	Advice that involves pesticides or chemicals shall include basic safety precautions and recommend consulting a local agricultural extension officer.

3.8 Compatibility and Maintainability

ID	Requirement
NFR-27	The web application shall work on the latest two versions of Google Chrome, Microsoft Edge, Mozilla Firefox and Safari, and on Chrome for Android.
NFR-28	The source code shall be maintained in a Git repository with a clear folder structure, and the REST API shall be documented.
NFR-29	The system shall log errors on the server to support debugging and maintenance.

4. User Roles

The following users and external systems interact with KrishiBondhu AI.

4.1 Farmer / User

The primary user of the system. This role covers farmers, agricultural students and members of rural communities who use the platform to obtain agricultural guidance.

Responsibilities:

- Register, log in and manage a personal profile and language preference.
- Use the system features: AI assistant, crop disease detection, fertilizer guidance, crop recommendations, weather and pest information.
- Interact with the AI components and give feedback on AI responses.
- Manage personal data: farm records, reminders and saved recommendations.

4.2 Agricultural Advisor

A trusted agricultural expert (for example, an agriculture graduate or extension worker) who helps keep the system's guidance accurate.

Responsibilities:

- Review AI responses flagged by users and submit corrections or verified guidance.
- Add and update pest and crop knowledge entries.
- Help maintain the quality and reliability of agricultural content.

4.3 Admin

The system administrator responsible for managing the platform.

Responsibilities:

- Manage users and assign roles (view, suspend, reactivate, delete accounts).
- Manage the pest and crop knowledge base.
- Monitor system activities through activity logs.
- Generate usage reports.

4.4 External Systems

- Google Gemini API: provides AI text responses and image analysis, invoked by the backend.
- Weather API: provides current weather and forecast data, invoked by the backend.

Note: A Moderator role (community content moderation) is not required because the current scope has no public community or user-generated discussion features. The Advisor role covers review of flagged AI content.

5. Use Cases

Each use case below shows the actor, the action and the system response. Use case IDs match the Use Case Diagram (Section 6).

UC-01: Register Account	
Actor	Farmer / User
Action	User opens the registration page and enters name, email and password (mobile number optional).
System Response	System validates the data, creates the account with a securely hashed password and confirms registration.
Alternative / Error	If the email already exists or the password is weak, the system shows an error and asks the user to correct the input.

UC-02: Login / Logout	
Actor	Farmer / User
Action	User enters email and password (or selects Logout).
System Response	System validates credentials and grants access to the dashboard (or ends the session on logout).
Alternative / Error	Invalid credentials: the system shows an error; repeated failures are rate limited.

UC-03: Manage Profile	
Actor	Farmer / User
Action	User opens the profile page and edits name, location (district/upazila), main crops and farm size.
System Response	System validates and saves the changes and shows a confirmation.
Alternative / Error	Invalid input: the system highlights the fields to correct.

UC-04: Select Language	
Actor	Farmer / User
Action	User chooses Bangla or English from the language switch.
System Response	System immediately displays the interface in the chosen language and stores the preference.
Alternative / Error	-

UC-05: Manage Farm Records	
Actor	Farmer / User
Action	User creates, views, edits or deletes a farm record (crop, field, area, soil type, sowing and harvest dates) and adds notes.
System Response	System stores the changes, updates the records list and confirms the action.
Alternative / Error	Missing required fields: the system shows validation errors.

UC-06: Manage Reminders	
Actor	Farmer / User
Action	User creates, edits, deletes or completes a reminder with a date, time and activity type.
System Response	System saves the reminder and shows an in-app notification when it becomes due.
Alternative / Error	Date in the past: the system asks the user to choose a valid time.

UC-07: Save & View Recommendations	
Actor	Farmer / User
Action	User selects Save on an AI recommendation, or opens the Saved Recommendations page to search or delete items.
System Response	System stores the recommendation in the user's saved list and displays or removes items as requested.
Alternative / Error	-

UC-08: Browse Pest Information	
Actor	Farmer / User
Action	User searches or browses the pest library by name or affected crop and opens a pest entry.
System Response	System displays the pest's description, symptoms, affected crops and management methods.
Alternative / Error	No match: the system shows a 'no results' message with search suggestions.

UC-09: Give Feedback on AI Response	
Actor	Farmer / User
Action	User marks an AI response as helpful or not helpful, or flags it as incorrect with an optional comment.
System Response	System records the feedback and queues flagged responses for advisor review.
Alternative / Error	-

UC-10: Ask AI Assistant	
Actor	Farmer / User; Google Gemini API
Action	User types a farming question in Bangla or English.
System Response	System sends the question with profile context to the Gemini API, displays the generated answer with an advisory notice and stores it in the chat history.
Alternative / Error	If the API is unavailable or times out, the system shows an error and offers a retry.

UC-11: Detect Crop Disease from Image	
Actor	Farmer / User; Google Gemini API
Action	User uploads or captures a photo of an affected crop.
System Response	System validates the image, sends it for analysis and displays the probable disease, confidence, symptoms and management steps; the result is saved to the scan history.
Alternative / Error	Unsupported/oversized file or unclear image: the system asks for a different image.

UC-12: Get Fertilizer Guidance	
Actor	Farmer / User; Google Gemini API
Action	User enters crop, soil type, growth stage and field area and submits the request.
System Response	System generates and displays general fertilizer guidance (type, quantity guidance, timing, safety notes) with the option to save it.
Alternative / Error	Incomplete input: the system highlights the missing fields.

UC-13: Get Crop Recommendation	
Actor	Farmer / User; Google Gemini API
Action	User provides soil type, season and location (or uses the profile location).
System Response	System returns suggested crops with short reasons for each suggestion.
Alternative / Error	-

UC-14: View Weather Updates	
Actor	Farmer / User; Weather API
Action	User opens the weather page.
System Response	System retrieves and displays current weather, a short forecast and farming advisories for the user's location.
Alternative / Error	Weather service unavailable: the system shows the last available data or an error message.

UC-15: Staff Login	
Actor	Advisor / Admin
Action	Advisor or Admin enters credentials on the login page.
System Response	System validates the credentials, checks the role and opens the corresponding staff dashboard.
Alternative / Error	Non-staff account or invalid credentials: access to staff pages is denied.

UC-16: Review Flagged AI Responses	
Actor	Advisor
Action	Advisor opens the list of flagged AI responses, reviews one and submits a correction or marks it as reviewed.
System Response	System stores the review, updates the status of the flagged item and logs the action.
Alternative / Error	-

UC-17: Manage Pest & Crop Knowledge Base	
Actor	Advisor / Admin
Action	Staff member adds, edits or removes a pest or crop entry.
System Response	System validates and saves the entry, which becomes available to users in the pest library.
Alternative / Error	Duplicate or invalid entry: the system shows an error.

UC-18: Manage Users & Roles	
Actor	Admin
Action	Admin searches for a user and suspends, reactivates, deletes the account or changes its role.
System Response	System applies the change, confirms it and records it in the activity log.
Alternative / Error	Admin attempts to remove their own admin role/account: the system blocks the action.

UC-19: Monitor System Activity	
Actor	Admin
Action	Admin opens the activity log and filters by user, action or date.
System Response	System displays matching log entries (login, deletion, role change, review events).
Alternative / Error	-

UC-20: Generate Usage Reports	
Actor	Admin
Action	Admin selects a report type and a date range.
System Response	System compiles and displays usage statistics (users, AI queries, disease scans) and allows download.
Alternative / Error	No data in range: the system shows an empty-report message.

6. Initial System Diagrams

The following initial diagrams accompany this SRS. Full-size versions are attached at the end of this document (Appendix A, Figures 1 to 3) and are also submitted as separate PDF files.

Diagram	Description	File
Figure 1: Use Case Diagram	Shows the actors (Farmer/User, Advisor, Admin, Gemini API, Weather API), their interactions and the 20 major use cases (UC-01 to UC-20) of the system.	CSE4104-7G-T03_UseCaseDiagram.pdf
Figure 2: ER Diagram	Shows 14 database entities, their relationships, primary keys (PK) and foreign keys (FK). Implemented as MongoDB collections.	CSE4104-7G-T03_ERDiagram.pdf
Figure 3: System Architecture Diagram	Shows the React frontend, Node.js/Express backend API, MongoDB database, image storage and the external Google Gemini, weather and email services.	CSE4104-7G-T03_ArchitectureDiagram.pdf

6.1 Main Entities (summary)

Entity	Purpose
User	Account, role (Farmer/Advisor/Admin), language preference and location.
ChatSession, ChatMessage	AI assistant conversations and their messages.
Feedback	User ratings/flags on AI messages and advisor review.
DiseaseScan	Uploaded crop image, AI diagnosis result and advice.
Recommendation, SavedRecommendation	AI recommendations (crop, fertilizer, disease, chat) and the ones users save.
Crop, Pest, CropPest	Knowledge base of crops and pests; CropPest resolves the many-to-many relationship.

FarmRecord, FarmNote	User's crop/field records and dated notes.
Reminder	Scheduled farming activities, optionally linked to a farm record.
ActivityLog	Audit trail of important actions for admin monitoring.

7. Consistency with the Project Proposal

The table below maps each feature planned in the project proposal / README to the requirements and use cases of this SRS.

Proposed Feature	Functional Requirements	Use Cases
AI Agricultural Assistant	FR-09 to FR-14	UC-10, UC-09
Crop Disease Detection	FR-15 to FR-19	UC-11
Fertilizer Guidance	FR-20, FR-21	UC-12
Crop Recommendations	FR-22, FR-23	UC-13
Weather Updates	FR-24, FR-25	UC-14
Pest Management	FR-26, FR-27, FR-38, FR-41	UC-08, UC-17
Farm Records	FR-28 to FR-30	UC-05
Reminders	FR-31 to FR-33	UC-06
Saved Recommendations	FR-34, FR-35	UC-07
Bangla and English support	FR-02, FR-07, FR-08	UC-04
Accounts and administration (supporting)	FR-01 to FR-06, FR-36 to FR-46	UC-01 to UC-03, UC-15 to UC-20

Disclaimer: KrishiBondhu AI provides preliminary, AI-generated agricultural guidance. It does not replace advice from qualified agricultural experts, and users should confirm important decisions (especially those involving chemicals or large investments) with a local agricultural extension officer.

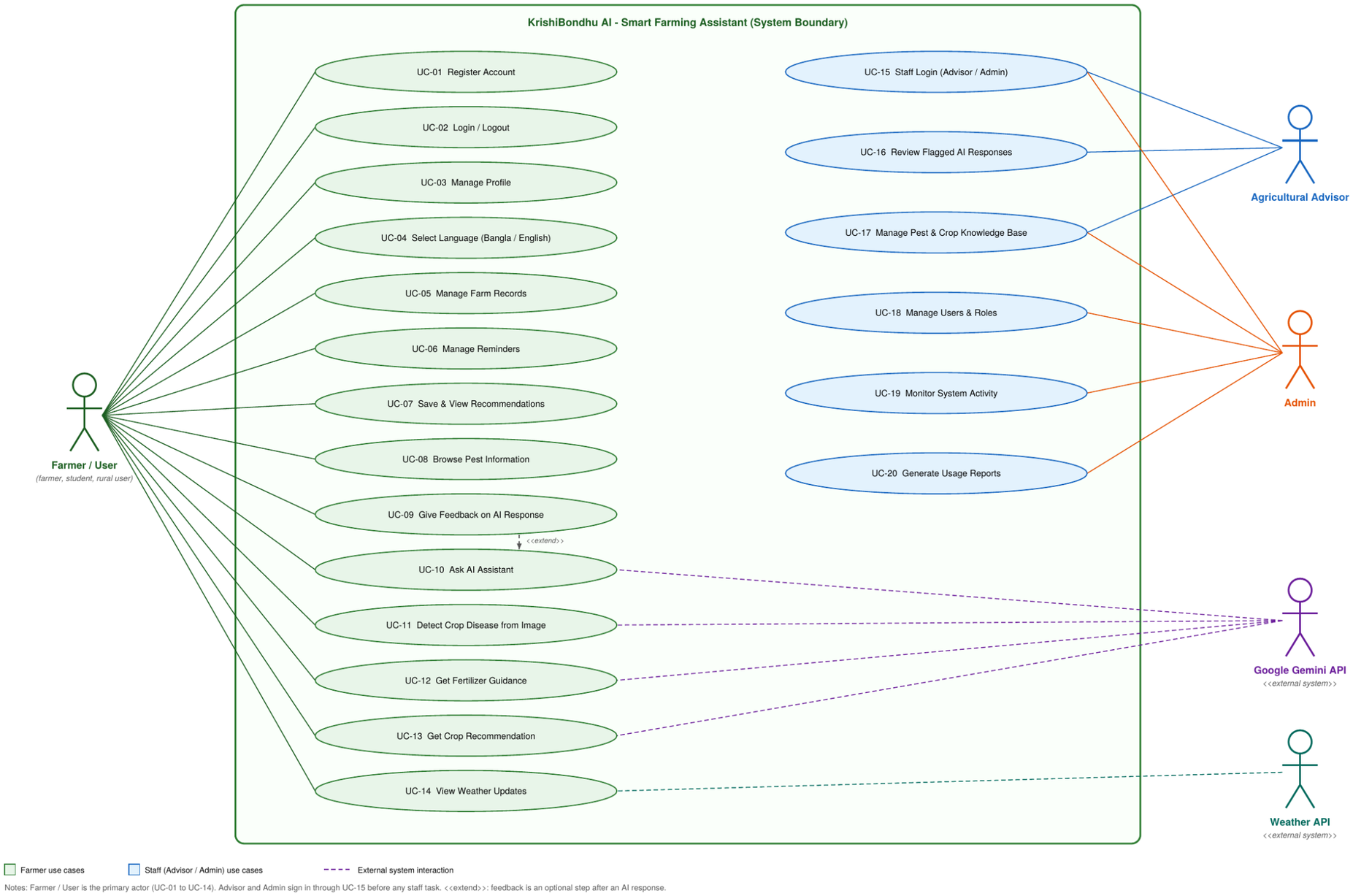


Figure 1: Use Case Diagram

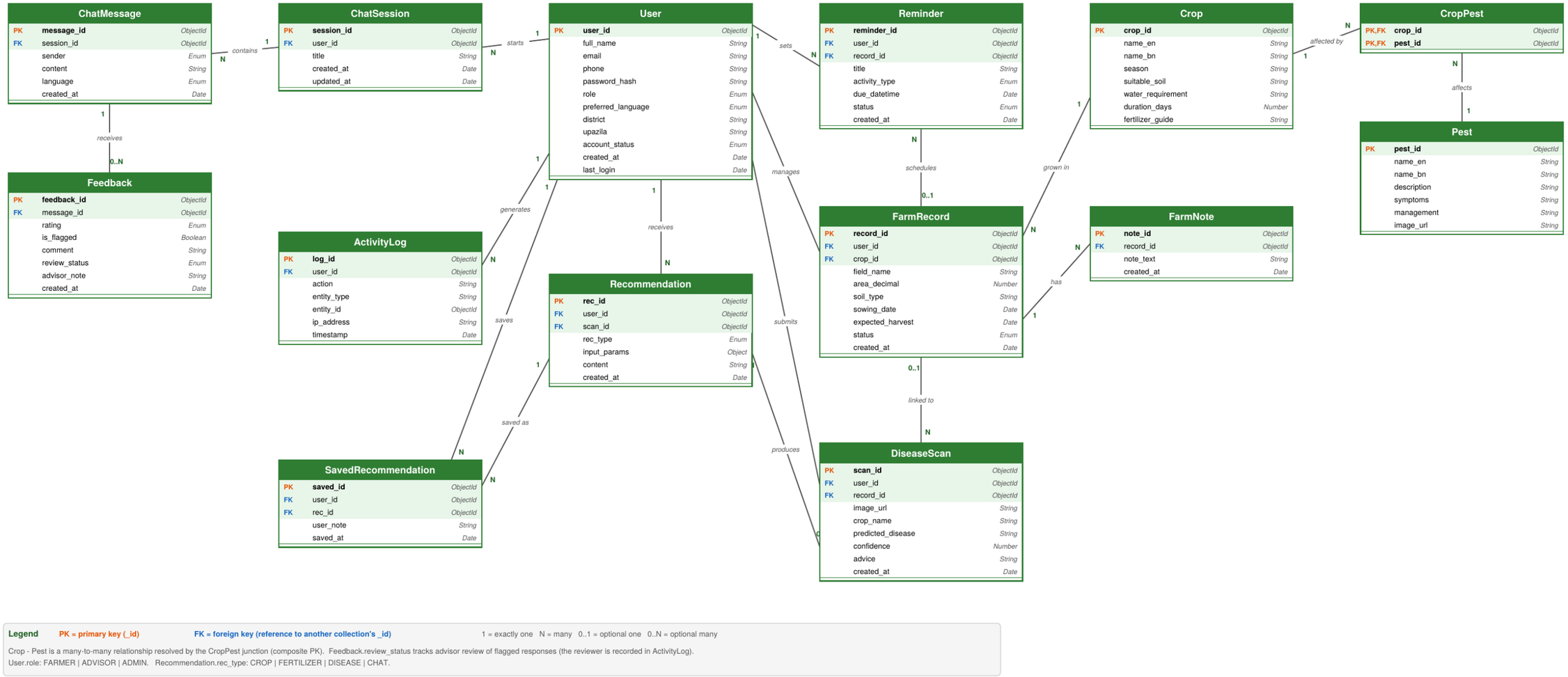


Figure 2: ER Diagram

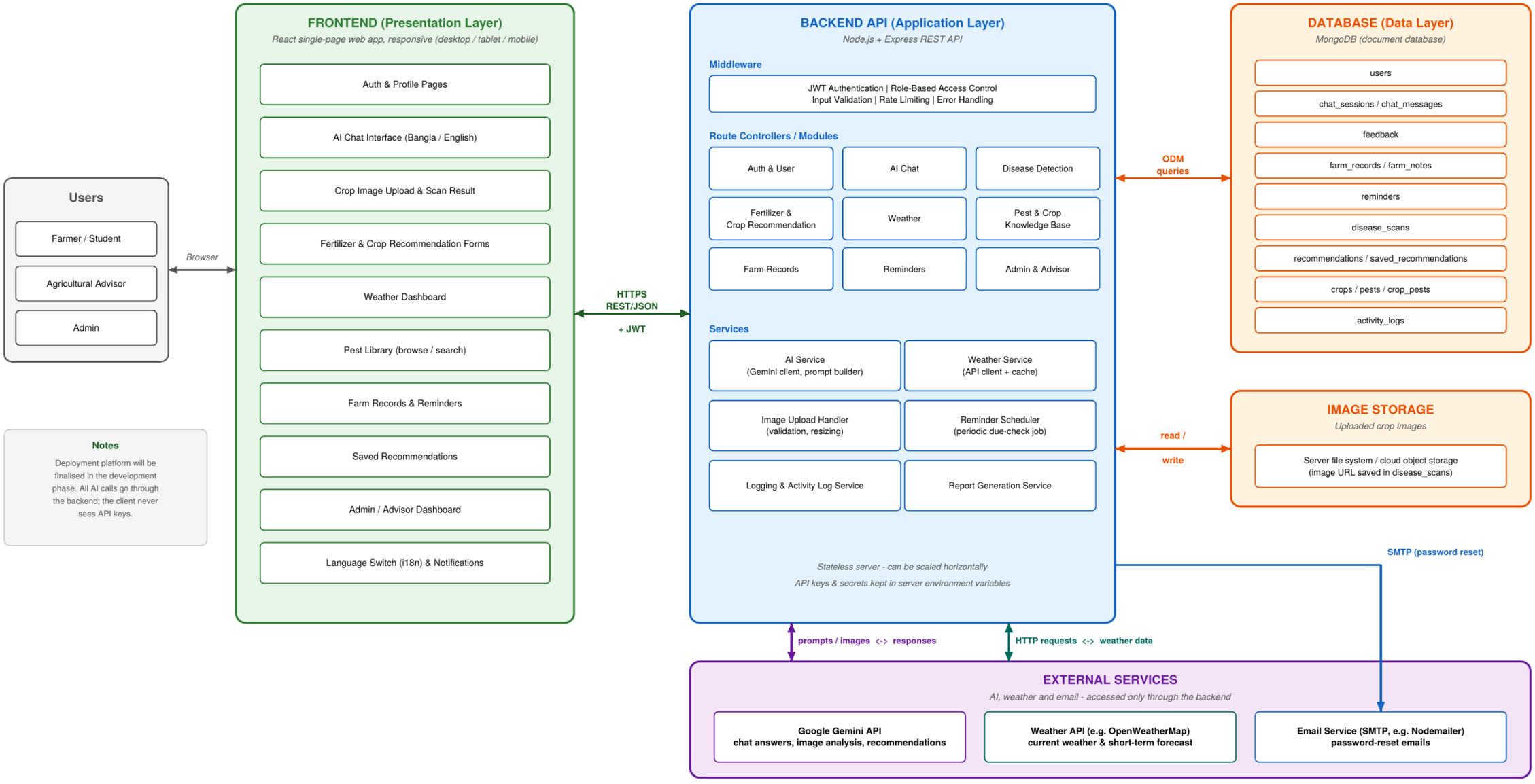


Figure 3: System Architecture Diagram